

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The time taken for half of the atoms of a radioactive isotope to decay (activity halves)	1
01.2	• $32 / 8 = 4$ half-lives • $6400 / 2^4 = 400$ Bq	2
01.3	• Short half-life isotopes (I-131) release all their radiation quickly: dangerous but only briefly so short-term precautions (potassium iodide tablets, evacuation) are sufficient • Very long half-life isotopes (U-238) emit only small amounts of radiation in any period so the level of danger is low • Decades-long half-lives combine a high rate of radiation release with persistence for most or all of a human lifetime, so short-term protection is inadequate (land contaminated for generations)	3
01.4	• Alpha particles have very low penetrating power and are absorbed by clothing / dead skin so external sources cause little exposure • Inside the body the alpha particles are absorbed in a very small mass of tissue so damage is concentrated (high RBE - weighting factor 20) • Long half-life means it continues to emit for the rest of the person's life / cannot be excreted quickly (contamination not just exposure)	3
01.5	• A previously stable nucleus that becomes radioactive after absorbing a neutron (eg in reactor structures), later decaying to emit ionising radiation	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 299-301; AQA spec 3.4.3.2.14.	
02.1	• Natural processes may concentrate the material (eg adsorption onto sediments, bioaccumulation and food-chain concentration) • It may be carried to locations where it is hazardous, such as fishing grounds or beaches used by people	2

Question	Marking guidance	Marks
02.2	Any 4 from: - Physical state of the effluent (solution / solid) - Ocean currents / river flow - Effect of pH and oxygen on solubility - Bioaccumulation and food-chain concentration - Food sources and consumption by local people (eg shellfish, seaweed) - Half-lives of the isotopes - Meteorological conditions for any gaseous release - Nature of the environment - geology, vegetation	4
02.3	- Monitoring focuses on the sites most at risk of contamination - No need to sample sites that were never likely to be polluted; sampling checks the accuracy of the predictions	2
02.4	- It assesses the risk to the people most exposed because of their lifestyle before any health effects occur - If their risk is acceptably low everyone else has a lower risk; if unacceptable, emissions are reduced to prevent future exposure	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
03.1	- Each regional population has its own unique gene pool / DNA profile - DNA from seized timber or ivory is compared with the database of regional populations - This identifies where the material came from, showing whether it was legally harvested and where poaching occurs	3
03.2	- Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) - Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions	2
03.3	- Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) - Drones: cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys; infrared cameras detect animals at night	3
03.4	- Cameras record automatically (visible light by day, infrared at night) without researchers being present - So animals behave naturally and nests are not deserted; monitoring continues 24 hours	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
04.1	As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures)	1
04.2	About 39 °C (accept 38-40)	1
04.3	9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) = 1.2 mg dm⁻³ (accept 1.1-1.3)	2
04.4	- Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line - Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible	2

Question	Marking guidance	Marks
04.5	<i>Any 3 from:</i> • Increased metabolic rate so more energy used / less surplus for growth • Eggs develop faster / hatch earlier (out of step with food supply) • Non-indigenous warm-water species may thrive and out-compete indigenous species • Toxicity of other pollutants is greater as toxins are metabolised faster • Reduced resistance to disease • Proteins / enzymes denatured if temperature is very high (above about 40 °C)	3
04.6	• Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	
05.1	• The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions • The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling	2
05.2	• Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) • Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced)	2
05.3	• Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
06.1	• Terracing: sloping land divided into narrow level fields held by retaining walls along contours; water slows as it crosses each terrace and infiltrates instead of running off • Tied ridging: on nearly flat land a criss-cross of intersecting ridges retains rainwater, increasing infiltration and preventing runoff • Windbreaks: hedgerows / tree rows reduce wind velocity (for up to 30 times their height) so the wind has less kinetic energy to carry particles; protect bare fields after harvest • Mulching: organic matter on the surface protects soil from raindrop impact and wind; decomposes to humus which binds particles; retains moisture • Strip cropping: strips of different crops harvested at different times (at 90° to the wind) so there is always a crop protecting part of the field; slows runoff	5
06.2	• No frequent replanting / soil disturbance (ploughing) that breaks up soil structure and exposes particles • Permanent root systems and canopy cover protect the soil all year	2
06.3	<i>Any 3 from:</i> • Control stocking density so vegetation regrows faster than it is eaten • Restrict access to vulnerable areas such as river banks (fencing) • Move / remove livestock at high-risk times, eg after heavy rain when soil is wet • Rotational grazing to rest pastures	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 164-166; AQA spec 3.2.5.3.	
07.1	- Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges - The particles clump together (flocs) and settle in the clarifier tank	3
07.2	- Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste	2
07.3	- Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution)	2
07.4	- Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) - Reverse osmosis / distillation (desalination) - expensive, energy-intensive - Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
08.1	- When there is a difference in level between the sea and the basin: on the ebb (about 6-8 hours) water flows out through the turbines - Also possible on the flood when the sea is higher than the basin; no generation when levels are equal (slack water)	2
08.2	- Two periods of slack water in each 12.5-hour cycle (twice in about 25 hours) give no flow; the timing does not match demand - Tides depend on the positions of the Moon and Sun which are known, so output can be forecast years ahead and alternative supplies planned	2
08.3	- Spring tides: Sun and Moon aligned so their gravitational effects combine giving a large tidal range; neap: effects oppose giving a small range - Energy depends on the mass of water and height difference, so output is much greater at springs	2
08.4	- When demand is low, surplus electricity pumps water into the lagoon above sea level - The stored water is released through the turbines later when demand is high (pumped storage)	2
08.5	- Large tidal range (eg Severn Estuary) - Estuaries, bays or headlands / channels that focus the flow and increase velocity	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 214-215; AQA spec 3.3.4.1.	
09.1	The dose that kills 50% of the (test) population	1

Question	Marking guidance	Marks
09.2	• Insecticide X: LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well • Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) • Y has a much higher LD50 for the beetle (150) so few predators killed / more specific	3
09.3	<i>Any 2 from:</i> • High water solubility so it is leached into rivers / groundwater / runoff • Toxic to aquatic organisms downstream • Mobile so it can affect areas away from the field • Washed off the crop quickly so more frequent application needed	2
09.4	• Long half-life = high persistence so it is not broken down quickly • So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble)	2
09.5	<i>Any 2 from:</i> • Absorbed and translocated within the plant so protects new growth • Not washed off by rain • Does not need to cover all leaf surfaces / less spray needed • Less spray drift so less harm to non-target species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Fossil fuel technologies: secondary / tertiary recovery raising recovery from 20% to 60%; directional drilling and subsea wells; ROVs; fracking for tight oil and gas; tar sands / oil shales; coal gasification / liquefaction; enhanced gas recovery; methane hydrates; CCS (pre / post combustion, storage) • Limits: still finite (discoveries declining since the 1980s); rising energy and financial cost of unconventional sources; environmental impacts (aquifers, water use, tremors, land take, habitat damage); CO₂ and other pollution; CCS unproven at scale and impractical for vehicles • Nuclear: new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash); molten salt reactors; plutonium breeder and thorium reactors increasing fuel use many-fold; fusion as a long-term prospect • Limits: cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium; fusion not commercial • Sustainability criteria: maintaining supply without unacceptable environmental, economic or social impact; comparison with renewables and energy conservation; energy gap argument • Judgement: technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely; role as transition / base-load alongside renewables	9
10.2	• 80-90%	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25